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Advanced Concepts of Modeling in AI (2025-26)

MULTIPLE CHOICE QUESTIONS (MCQ)

1. Which of the following is the umbrella term that covers both Machine Learning and Deep Learning?
(a) Machine Learning
(b) Deep Learning
(c) Artificial Intelligence
(d) Data Science
2. What is the primary characteristic of a Supervised Learning model?
(a) It works on unlabeled data.
(b) It learns by trial and error based on a reward mechanism.
(c) It requires labeled data for training.
(d) It is based on a predefined set of rules.
3. Which type of AI model is based on rules and instructions defined by a developer?
(a) Learning-Based Approach
(b) Reinforcement Learning
(c) Unsupervised Learning
(d) Rule-Based Approach
4. A model that predicts a continuous value like temperature or price is a:
(a) Classification Model
(b) Regression Model
(c) Clustering Model
(d) Association Model
5. What is the main advantage of Neural Networks mentioned in the document?
(a) They are based on a simple decision tree approach.
(b) They require minimal data for training.
(c) They are able to automatically extract data features.
(d) They only work on numerical data.
6. The process of attaching meaning or tags to data is known as:
(a) Data Processing
(b) Data Mining
(c) Data Labeling
(d) Data Extraction
7. Which of the following is a subset of Machine Learning?
(a) Artificial Intelligence
(b) Supervised Learning
(c) Rule-Based Approach
(d) All of the above
8. In an Artificial Neural Network, which layer is responsible for processing the data using weights and biases?
(a) Input Layer
(b) Hidden Layer
(c) Output Layer

- (d) All layers
9. A supermarket uses an AI model to group its customers based on their purchase history to send targeted offers. What kind of model is this?
- (a) Supervised Learning
 - (b) Reinforcement Learning
 - (c) Classification
 - (d) Clustering
10. Which learning approach is characterized by machines learning from feedback in a trial-and-error method?
- (a) Supervised Learning
 - (b) Unsupervised Learning
 - (c) Reinforcement Learning
 - (d) Classification
11. The core processing in an Artificial Neural Network occurs in which layer?
- (a) Input Layer
 - (b) Hidden Layer
 - (c) Output Layer
 - (d) All layers
12. A model is trained to classify handwritten digits (0-9). This is an example of:
- (a) Unsupervised Learning
 - (b) Regression
 - (c) Classification
 - (d) Clustering
13. Which of the following is a drawback of the Rule-Based Approach?
- (a) It requires vast amounts of data.
 - (b) It is too complex to implement.
 - (c) The learning is static and does not adapt to changes.
 - (d) It cannot be used for chatbots.
14. In the context of data, what are "features"?
- (a) Rows of a table
 - (b) The output of a model
 - (c) Columns of a table
 - (d) The final prediction
15. What is the difference between a training dataset and a testing dataset?
- (a) Training data is unlabeled, while testing data is labeled.
 - (b) Training data is used to teach the model, and testing data is used to evaluate its accuracy.
 - (c) Training data contains only features, and testing data contains only labels.
 - (d) There is no difference; they are interchangeable.
16. A model that predicts whether an email is "spam" or "not spam" is a type of:
- (a) Regression model
 - (b) Clustering model
 - (c) Classification model
 - (d) Association model
17. In the example of predicting a coin's currency based on its weight, what is the 'feature'?
- (a) The weight

- (b) The currency
 - (c) The coin's name
 - (d) The number of coins
18. An AI model that discovers patterns in an unlabeled dataset of dog images, such as clustering them by color or size, uses which approach?
- (a) Supervised Learning
 - (b) Reinforcement Learning
 - (c) Unsupervised Learning
 - (d) Rule-Based Approach
19. Which of the following is a characteristic of a Learning-Based Approach?
- (a) It is based on static rules.
 - (b) It learns from explicit programming.
 - (c) It adapts to changes in data.
 - (d) It does not require any data.
20. What is a "perceptron"?
- (a) A type of neural network layer
 - (b) A machine learning algorithm
 - (c) A simplified model of how an AI makes a decision
 - (d) A type of data
21. Which type of model is used to predict a car's approximate selling price based on parameters like fuel type and years of service?
- (a) Classification
 - (b) Regression
 - (c) Clustering
 - (d) Association
22. What is the role of the output layer in an Artificial Neural Network?
- (a) To process data and pass it to the hidden layers.
 - (b) To perform calculations with weights and biases.
 - (c) To present the final processed data to the user.
 - (d) To acquire data and feed it to the network.
23. In the context of a supermarket, which unsupervised learning method would be used to find a relationship between customers buying bread and also buying butter?
- (a) Classification
 - (b) Clustering
 - (c) Regression
 - (d) Association
24. Which type of learning is analogous to a child learning to swim on his own without a teacher?
- (a) Supervised Learning
 - (b) Unsupervised Learning
 - (c) Reinforcement Learning
 - (d) Rule-Based Learning
25. What do you need to train a supervised learning model?
- (a) Unlabeled data
 - (b) Only features
 - (c) Only labels
 - (d) Labeled data

26. What is the main drawback of a rule-based AI model?
- (a) It is too expensive to implement.
 - (b) It fails to learn from its mistakes.
 - (c) It requires continuous human intervention.
 - (d) It can only handle a single type of data.
27. When an AI model trains itself to perform tasks with vast amounts of data, it falls under which category?
- (a) Machine Learning
 - (b) Deep Learning
 - (c) Rule-Based Approach
 - (d) Supervised Learning
28. Which layer in an Artificial Neural Network does not perform any processing?
- (a) Input layer
 - (b) Hidden layer
 - (c) Output layer
 - (d) Both (a) and (c)
29. Object classification in deep learning uses powerful algorithms to identify and label objects within:
- (a) An image
 - (b) A spreadsheet
 - (c) A text document
 - (d) A database
30. The process of a neural network finding the right output by adjusting weights based on the error is known as:
- (a) Trial and error
 - (b) Backpropagation
 - (c) Forward propagation
 - (d) Activation
31. What is the key difference between Clustering and Classification?
- (a) Classification uses unlabeled data, while clustering uses labeled data.
 - (b) Classification assigns objects to predefined classes, while clustering finds similarities and groups objects.
 - (c) Classification is a supervised model, while clustering is a reinforcement model.
 - (d) Clustering requires more data than classification.
32. What is the goal of Reinforcement Learning?
- (a) To identify relationships in unlabeled data.
 - (b) To predict a continuous value.
 - (c) To make a series of decisions that maximize a reward.
 - (d) To classify data into discrete categories.
33. Anomaly detection, such as flagging a sudden spike in a heart rate, is an example of which type of model?
- (a) Classification
 - (b) Regression
 - (c) Machine Learning
 - (d) Reinforcement Learning

34. In the example of a spam email filter, what serves as the 'labels' during the training phase?
- (a) The words in the email
 - (b) The sender's information
 - (c) The classification of emails as either "spam" or "legitimate"
 - (d) The email attachments
35. What is the purpose of a testing dataset?
- (a) To train the model with new data.
 - (b) To check for errors in the code.
 - (c) To evaluate the accuracy of the trained model.
 - (d) To find hidden patterns.
36. Which of the following is an example of a Classification problem?
- (a) Predicting the price of a house.
 - (b) Predicting the number of days a patient will stay in a hospital.
 - (c) Predicting whether a patient will have a short or long hospital stay.
 - (d) Predicting a city's average temperature for the next month.
37. What type of data is used for training a Regression model?
- (a) Categorical data
 - (b) Discrete data
 - (c) Continuous data
 - (d) Unlabeled data
38. Which type of model is used by OTT platforms like Netflix to recommend movies based on a user's watch history?
- (a) Supervised Learning
 - (b) Regression
 - (c) Clustering
 - (d) Reinforcement Learning
39. The structure of an Artificial Neural Network is inspired by:
- (a) The human brain and nervous system
 - (b) The human digestive system
 - (c) A computer's hardware components
 - (d) A mathematical formula
40. In a supervised learning example, what does a model learn from the training data?
- (a) To create new features
 - (b) To identify new patterns without guidance
 - (c) To apply the knowledge to test data
 - (d) To define the rules itself
41. Which of the following is a primary type of AI model mentioned in the document?
- (a) Predictive-based
 - (b) Reward-based
 - (c) Learning-based
 - (d) Both (b) and (c)
42. What is a "label" in the context of data?
- (a) A column of a table
 - (b) The name of the dataset
 - (c) A tag that gives meaning to data
 - (d) The algorithm used in the model

43. A Convolutional Neural Network (CNN) is a type of:
- (a) Machine Learning model
 - (b) Rule-Based model
 - (c) Deep Learning algorithm
 - (d) Reinforcement Learning model
44. When a model predicts a discrete value, such as "hot" or "cold" weather, it is using a:
- (a) Regression model
 - (b) Classification model
 - (c) Association model
 - (d) Clustering model
45. The "funnel type approach" described for AI, ML, and DL suggests that:
- (a) DL has more applications than AI.
 - (b) All ML applications are also DL applications.
 - (c) All AI applications are also ML applications.
 - (d) DL has a very specific set of applications, a subset of ML, which is a subset of AI.
46. Which learning approach would be used to analyze bank data for suspicious transactions without a predefined definition of what is "suspicious"?
- (a) Supervised Learning
 - (b) Unsupervised Learning
 - (c) Regression
 - (d) Classification
47. What is a key function of the hidden layer in a neural network?
- (a) To acquire input data.
 - (b) To give the final output.
 - (c) To perform computations.
 - (d) To act as a user interface.
48. What is the main difference between a rule-based approach and a learning-based approach?
- (a) Rule-based is static, while learning-based is adaptive.
 - (b) Rule-based is more efficient for large datasets.
 - (c) Learning-based is based on predefined rules.
 - (d) There is no significant difference.
49. What are "weights" and "biases" used for in a neural network?
- (a) To define the final output.
 - (b) To acquire input data.
 - (c) To perform computations within the hidden layers.
 - (d) To create a user interface.
50. What does Deep Learning enable software to do?
- (a) To learn from a minimal amount of data.
 - (b) To train itself to perform tasks with vast amounts of data.
 - (c) To follow a predefined set of rules.
 - (d) To predict discrete values only.
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DETAILED ANSWERS FOR MCQS

1. (c) Artificial Intelligence
Artificial Intelligence is an umbrella term that includes both Machine Learning and Deep Learning as its subsets.
2. (c) It requires labeled data for training.
In Supervised Learning, the dataset fed to the machine is labeled. The model learns from this labeled data to predict future outcomes.
3. (d) Rule-Based Approach
The Rule-Based Approach involves a developer defining the rules or instructions that the machine follows to perform its task.
4. (b) Regression Model
A Regression model is a type of supervised learning model that works on continuous data and predicts a continuous output value.
5. (c) They are able to automatically extract data features.
The key advantage of neural networks is their ability to automatically extract data features without the need for programmer input.
6. (c) Data Labeling
Data Labeling is defined as the process of attaching meaning to data, which helps in identifying features and labels.
7. (b) Supervised Learning
Machine Learning can be further divided into three parts: Supervised Learning, Unsupervised Learning, and Reinforcement Learning.
8. (b) Hidden Layer
The hidden layers are where the core processing occurs. They perform computations using weights and biases on each node.
9. (d) Clustering
Clustering is an unsupervised learning method that groups data points (customers) into clusters based on similarities or patterns (purchase history) without having predefined labels.
10. (c) Reinforcement Learning
Reinforcement Learning is a type of learning where a machine learns to perform a task through a repeated trial-and-error method, with feedback provided to optimize decisions for a reward.
11. (b) Hidden Layer
The hidden layers in an Artificial Neural Network are where all the processing takes place. They are called 'hidden' because they are not visible to the user.
12. (c) Classification
This is a classification problem because the model is trained to assign data to predefined categories or classes (the digits 0-9).
13. (c) The learning is static and does not adapt to changes.
A drawback of the rule-based approach is that the learning is static. The model, once trained, does not adapt or learn from changes in the data.

14. (c) Columns of a table
"features" is the columns of a table containing information.
15. (b) Training data is used to teach the model, and testing data is used to evaluate its accuracy.
A training dataset is a collection of examples given to the model to learn from. A testing dataset is used to evaluate the accuracy of the model after it has been trained.
16. (c) Classification model
Classifying an email as "spam" or "not spam" involves assigning it to one of two discrete categories. This is a classic example of a classification problem.
17. (a) The weight
In this example, the weight of the coin is the input variable used to predict the currency. It states that the feature is the "weights".
18. (c) Unsupervised Learning
Unsupervised learning models work on unlabeled data to identify relationships, patterns, and trends. Clustering images based on color or size without prior labels is a perfect example of this.
19. (c) It adapts to changes in data.
Unlike rule-based models, learning-based models are adaptive. They get trained on data and can modify their algorithms to handle changes and exceptions.
20. (c) A simplified model of how an AI makes a decision
The term "perceptron" used to illustrate a simplified model of how an AI makes a decision by summing up weighted inputs, adding a bias, and comparing the result to a threshold.
21. (b) Regression
The model predicts a continuous value (an approximate price) based on various parameters. This makes it a regression problem.
22. (c) To present the final processed data to the user.
The output layer receives the final processed data from the last hidden layer and gives it to the user as the final output. It does not process the data itself.
23. (d) Association
An association rule is an unsupervised learning method used to find interesting relationships between variables. The example of customers buying bread and butter is a typical association rule problem.
24. (b) Unsupervised Learning
The analogy of a child learning is to swim on his own without any supervision to explain the concept of Unsupervised Learning.
25. (d) Labeled data
Supervised learning models are trained using labeled data, where each piece of data is tagged with some information.
26. (b) It fails to learn from its mistakes.
A major drawback is that a rule-based model does not improvise itself based on feedback and will fail if tested on a dataset different from the rules it was originally fed.
27. (b) Deep Learning

Deep learning enables software to train itself to perform tasks with vast amounts of data.

28. (d) Both (a) and (c)

The input layer's job is to acquire data, and the output layer's job is to present the final output. Neither of these layers performs any processing.

29. (a) An image

Object identification/classification in deep learning is described as the task of identifying and labeling objects within an image.

30. (a) Trial and error

The process of a neural network is finding the right output as a trial-and-error process where weights are adjusted based on the error between the desired output and the network output.

31. (b) Classification assigns objects to predefined classes, while clustering finds similarities and groups objects.

Classification uses predefined classes, while clustering finds similarities to group objects into clusters.

32. (c) To make a series of decisions that maximize a reward.

Reinforcement learning is a learning approach that enables a computer to make a series of decisions to maximize a reward metric without direct programming.

33. (c) Machine Learning

Anomaly detection is listed as a prime example of a Machine Learning application.

34. (c) The classification of emails as either "spam" or "legitimate"

In the spam email filter example, the model is provided with a large dataset of emails, each labeled as either "spam" or "legitimate" during the training phase. These are the labels.

35. (c) To evaluate the accuracy of the trained model.

The testing dataset is used to test the accuracy of the model after it has been trained.

36. (c) Predicting whether a patient will have a short or long hospital stay.

This is a classification problem because the output is a discrete category (short or long), not a continuous value.

37. (c) Continuous data

Regression models work on continuous data, such as temperature, price, or income.

38. (c) Clustering

Clustering techniques are used by OTT platforms like Netflix and Spotify to recommend items based on similarities in user preferences.

39. (a) The human brain and nervous system

Artificial Neural Networks are "loosely modelled after how neurons in the human brain behave" and the human nervous system.

40. (c) To apply the knowledge to test data

In supervised learning, the model learns from the training data and then applies that knowledge to the test data to predict the output.

41. (c) Learning-based

AI models classified into two main categories: Rule-based and Learning-based.

42. (c) A tag that gives meaning to data
A label is "some information which can be used as a tag for data".
43. (c) Deep Learning algorithm
Convolutional Neural Network (CNN) is a Deep Learning algorithm.
44. (b) Classification model
A classification model is used to predict a discrete value, such as "hot" or "cold" weather.
45. (d) DL has a very specific set of applications, a subset of ML, which is a subset of AI.
This is a "funnel type approach" where AI is the largest, followed by ML, and then DL, which has a very specific and limited set of applications within ML.
46. (b) Unsupervised Learning
Since the suspicious transactions are not defined, the model would be an unsupervised learning model that identifies outliers or anomalous transactions.
47. (c) To perform computations.
The hidden layers in a neural network are where the processing occurs, performing computations using weights and biases.
48. (a) Rule-based is static, while learning-based is adaptive.
Rule-based learning is static and does not adapt, whereas learning-based models can modify themselves according to changes in data.
49. (c) To perform computations within the hidden layers.
Weights and biases are used in the hidden layers to perform computations on the data received from the input layer.
50. (b) To train itself to perform tasks with vast amounts of data.
Deep learning "enables software to train itself to perform tasks with vast amounts of data".
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ASSERTION AND REASON QUESTIONS

Directions: Choose the correct option from the following:

- (a) Both A and R are true, and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true, but R is false.
- (d) A is false, but R is true.

1. **Assertion (A):** Supervised learning models require labeled data for training.
Reason (R): Labeled data acts as a guide, providing the model with a clear relationship between features and outcomes.
2. **Assertion (A):** A model predicting a house price based on its size is a Regression model.
Reason (R): Regression models are used to predict discrete values.
3. **Assertion (A):** Anomaly detection is a key application of Unsupervised Learning.

- Reason (R):** Anomaly detection models look for outliers and unusual patterns in unlabeled data.
4. **Assertion (A):** A rule-based AI model is adaptive and can handle new exceptions it was not explicitly programmed for.
Reason (R): Rule-based models follow a static set of instructions predefined by a developer.
 5. **Assertion (A):** Deep Learning is a subset of Machine Learning.
Reason (R): Deep Learning algorithms are based on artificial neural networks and require vast amounts of data for self-training.
 6. **Assertion (A):** Reinforcement Learning models learn by trial and error to maximize a reward.
Reason (R): This learning approach is beneficial for complex problems where pre-existing data is insufficient.
 7. **Assertion (A):** A model classifying an email as 'spam' or 'not spam' is a Classification model.
Reason (R): Classification models predict a continuous value.
 8. **Assertion (A):** The hidden layers of a neural network are where the core processing of data takes place.
Reason (R): The input layer and output layer of an ANN perform all the computational tasks.
 9. **Assertion (A):** Unsupervised learning is analogous to a teacher-student relationship.
Reason (R): Unsupervised learning models work on unlabeled data without any guidance.
 10. **Assertion (A):** The primary purpose of the testing dataset is to train the model.
Reason (R): The training dataset is used to evaluate the accuracy of the model.
 11. **Assertion (A):** The 'weights' in a neural network are adjusted during the learning process.
Reason (R): The adjustment of weights helps to reduce the error and find the right output.
 12. **Assertion (A):** Reinforcement learning is a type of supervised learning.
Reason (R): It involves a reward system that provides feedback, which is a form of supervision.
 13. **Assertion (A):** Clustering is a type of unsupervised learning.
Reason (R): It is used to group data points into clusters based on their similarities without using predefined classes.
 14. **Assertion (A):** A model predicting whether a patient will have a short or long hospital stay is a Regression model.
Reason (R): The output of this prediction is a continuous value.
 15. **Assertion (A):** The input layer of an Artificial Neural Network (ANN) does not process data.
Reason (R): Its sole function is to acquire data and feed it to the next layer.
 16. **Assertion (A):** An Association model finds relationships between variables in a dataset.
Reason (R): An example is recommending bread to a customer who buys butter.
 17. **Assertion (A):** Deep Learning requires only a small amount of data to train a model effectively.
Reason (R): Deep learning enables software to train itself with vast amounts of data.

18. **Assertion (A):** A perceptron is a simplified model of how an AI makes a decision.
Reason (R): It works by summing weighted inputs and a bias, then comparing the result to a threshold.
19. **Assertion (A):** A spam email filter is a good example of a learning-based model.
Reason (R): It adapts and improves its accuracy over time as it encounters new types of spam.
20. **Assertion (A):** The process of attaching meaning or tags to data is called Data Labeling.
Reason (R): Data Labeling is a crucial step in preparing data for Unsupervised Learning.

DETAILED ANSWERS

1. **Correct Answer:** (a)
Detailed Answer: Both statements are true. **Assertion (A)** is correct because supervised learning models are, by definition, trained on data that is already labeled. **Reason (R)** provides the correct explanation, as the labels serve as the "answer key" or guidance for the model to learn the correct relationship between input features and the desired output.
2. **Correct Answer:** (c)
Detailed Answer: **Assertion (A)** is true. Predicting a house price is a classic example of a regression problem because the output is a continuous numerical value (the price). However, **Reason (R)** is false. Regression models are used to predict **continuous** values, not discrete ones.
3. **Correct Answer:** (a)
Detailed Answer: Both statements are true, and the reason correctly explains the assertion. **Assertion (A)** is correct; anomaly detection is a key application of unsupervised learning because there are no predefined "labels" for what constitutes an anomaly. **Reason (R)** is also correct and provides the logical explanation: the model identifies these anomalies by independently finding unusual patterns or outliers in the unlabeled data.
4. **Correct Answer:** (d)
Detailed Answer: **Assertion (A)** is false. A rule-based AI model is not adaptive and cannot handle new exceptions it was not explicitly programmed for. It is static. **Reason (R)** is true, as the core principle of a rule-based model is to follow a static set of instructions predefined by the developer.
5. **Correct Answer:** (a)
Detailed Answer: Both statements are true, and the reason correctly explains the assertion. **Assertion (A)** is correct; the document clearly states that Deep Learning is a specific subset of Machine Learning. **Reason (R)** provides the correct explanation for this, highlighting that DL's core mechanism is its use of ANNs and its need for vast data, distinguishing it as a specific approach within the broader field of ML.
6. **Correct Answer:** (a)
Detailed Answer: Both statements are true, and the reason correctly explains the assertion. **Assertion (A)** is correct; Reinforcement Learning is fundamentally a trial-and-error process with the goal of maximizing a reward. **Reason (R)** correctly explains why this approach is so valuable: it can be applied to complex problems where it is difficult or impossible to create a pre-labeled dataset for training.

7. **Correct Answer:** (c)
Detailed Answer: **Assertion (A)** is true. Classifying an email into 'spam' or 'not spam' is a classic example of a classification problem because the output is one of two discrete categories. **Reason (R)** is false. Classification models predict discrete or categorical values, not continuous ones.
8. **Correct Answer:** (c)
Detailed Answer: **Assertion (A)** is true. The hidden layers of an ANN are where the core computations using weights and biases occur. **Reason (R)** is false. The input and output layers do not perform any processing; their roles are to acquire and present data, respectively.
9. **Correct Answer:** (d)
Detailed Answer: **Assertion (A)** is false. Supervised learning, not unsupervised learning, is analogous to a teacher-student relationship because the model learns from labeled data (the "teacher"). **Reason (R)** is true. Unsupervised learning models indeed work on unlabeled data and find patterns without any guidance.
10. **Correct Answer:** (d)
Detailed Answer: **Assertion (A)** is false. The testing dataset is used to **evaluate** the model's accuracy, not to train it. **Reason (R)** is also false. The **training** dataset is used to train the model, not to evaluate it.
11. **Correct Answer:** (a)
Detailed Answer: Both statements are true, and the reason correctly explains the assertion. **Assertion (A)** is correct, as the weights in a neural network are dynamic and adjusted during the training phase. **Reason (R)** correctly explains the purpose of this adjustment: to minimize the error between the model's prediction and the actual label, thereby improving accuracy.
12. **Correct Answer:** (c)
Detailed Answer: **Assertion (A)** is false. Reinforcement Learning is its own distinct type of learning and is not a subset of supervised learning. **Reason (R)** is true in that the feedback system is a form of learning, but it is not the same as the labeled guidance found in supervised learning.
13. **Correct Answer:** (a)
Detailed Answer: Both statements are true, and the reason correctly explains the assertion. **Assertion (A)** is correct; clustering is a fundamental technique within unsupervised learning. **Reason (R)** correctly describes the purpose and mechanism of clustering: to group similar data points without any predefined categories, which is the core principle of unsupervised learning.
14. **Correct Answer:** (d)
Detailed Answer: **Assertion (A)** is false. Predicting whether a hospital stay will be "short" or "long" is a classification problem because the output is a discrete category. **Reason (R)** is true in that it correctly states that the output for this specific problem would be a categorical value, not a continuous one.
15. **Correct Answer:** (a)
Detailed Answer: Both statements are true, and the reason correctly explains the assertion. **Assertion (A)** is correct, as the input layer is a passive conduit for data. **Reason (R)** provides the exact function of the input layer as described in the document, which is to acquire and feed the data without any processing.
16. **Correct Answer:** (c)

Detailed Answer: **Assertion (A)** is true, as an Association model is designed to find relationships between variables. **Reason (R)** is false. The example provided is incorrect; an association rule would be recommending **butter** to a customer who buys bread, not the other way around, as bread is the item that is known to have been purchased.

17. **Correct Answer:** (d)

Detailed Answer: **Assertion (A)** is false. The document explicitly states that Deep Learning requires **vast** amounts of data, not a small amount. **Reason (R)** is true and correctly describes the core function of Deep Learning.

18. **Correct Answer:** (a)

Detailed Answer: Both statements are true, and the reason correctly explains the assertion. **Assertion (A)** is correct; the perceptron is a simple, foundational model for how an AI makes a decision. **Reason (R)** provides the correct mechanism of how a perceptron works, which is the exact process of the simplified decision-making model.

19. **Correct Answer:** (a)

Detailed Answer: Both statements are true, and the reason correctly explains the assertion. **Assertion (A)** is correct; a spam email filter is a classic example of a learning-based model that learns from new data. **Reason (R)** correctly explains why it's an excellent example: it adapts to new data and improves its accuracy over time, a key characteristic of learning-based models.

20. **Correct Answer:** (c)

Detailed Answer: **Assertion (A)** is true. The process of attaching meaning or tags to data is indeed called Data Labeling. **Reason (R)** is false. Data labeling is a crucial step for **Supervised** Learning, not Unsupervised Learning. Unsupervised Learning works on unlabeled data.

2 MARKS QUESTIONS AND ANSWERS

1. Differentiate between a Rule-Based Approach and a Learning-Based Approach.

Ans. **A Rule-Based Approach** is where a developer explicitly defines the rules and instructions for the machine to follow. The learning is static, and the model does not adapt to new data. A **Learning-Based Approach**, on the other hand, is where the machine learns to perform a task by analyzing examples and finding patterns or rules on its own. These models are adaptive to changes in data.

2. What are the three main types of learning-based models?

Ans. The three main types of learning-based models are **Supervised Learning**, which uses labeled data; **Unsupervised Learning**, which works on unlabeled data to find patterns; and **Reinforcement Learning**, which learns through a reward mechanism.

3. What is the primary difference between labeled and unlabeled data?

Ans. Labeled data has a tag or label attached to it, providing meaning. Unlabeled data is the raw form of data with no tags attached.

4. Explain the concept of "features" and "labels" with an example.

Ans. Features are the columns of a dataset, describing the data. A label is a special feature that acts as a tag and is the target variable to be predicted. For example, if you are predicting the type of fruit based on its color, the color is the feature, and the fruit name is the label.

5. What is the role of a training dataset and a testing dataset?

Ans. A training dataset is a collection of examples used to teach or train an AI model. A testing dataset is then used to evaluate the accuracy and performance of the trained model.

6. Define the term "Artificial Neural Network" (ANN).

Ans. An Artificial Neural Network (ANN) is a system of machine learning algorithms modeled after the human brain and nervous system. Its key advantage is the ability to automatically extract data features without programmer input.

7. What are the two sub-categories of Supervised Learning?

Ans. The two sub-categories of Supervised Learning are Classification models, which predict discrete or categorical outputs, and Regression models, which predict continuous values.

8. Give one example of a Classification problem and one of a Regression problem.

Ans. A Classification problem example is predicting whether an email is "spam" or "not spam". A Regression problem example is predicting the price of a house based on its features.

9. Explain the concept of Unsupervised Learning with a simple analogy.

Ans. Unsupervised Learning is a type of learning without any guidance. An analogy is a child learning to swim on his own without any supervision, where the child discovers ways to swim by themselves, just as the model discovers patterns in unknown data.

10. What is a "perceptron," and what does it represent?

Ans. A perceptron is a simplified model of how an AI makes a decision. It takes multiple inputs, each with a weight, sums them up, adds a bias, and then compares the result to a threshold to produce an output.

11. In an Artificial Neural Network, what is the role of the input and output layers?

Ans. The input layer acquires data and feeds it into the network. The output layer receives the final processed data and presents it to the user as the final output. Neither of these layers performs any processing.

12. How does an AI model make a decision?

Ans. An AI model makes a decision by taking various factors as inputs, assigning a weight to each factor based on its importance, summing them up with a bias, and then comparing the result to a threshold to determine the output.

13. What is a key advantage of the Learning-Based Approach over the Rule-Based Approach?

Ans. A key advantage of the Learning-Based Approach is that the model is adaptive; it can modify itself and learn from new data, whereas a Rule-Based model is static and will not learn from its mistakes if the data or rules change.

14. What is the primary function of Unsupervised Learning models?

Ans. Unsupervised learning models are used to identify relationships, patterns, and trends from an unlabeled dataset. They help the user understand what the data is about by finding similarities and differences on their own.

15. What is a "label" in the context of supervised learning?

Ans. In supervised learning, a label is a piece of information or a tag that is attached to a data point to give it meaning. It is the known target that the model is being trained to predict.

16. Define Deep Learning.

Ans. Deep Learning is a specific learning approach and a subset of Machine Learning. It enables software to train itself to perform tasks with vast amounts of data by using multiple machine learning algorithms working together.

17. Give an example of an Association model in unsupervised learning.

Ans. An Association model is used to find relationships between variables. An example is a supermarket analyzing customer purchasing patterns and discovering that customers who buy bread are also likely to buy butter, which can be used to recommend items.

18. What is the key difference between a Classification model and a Clustering model?

Ans. A Classification model assigns data to predefined classes, while a Clustering model finds similarities between objects and groups them into clusters without any predefined classes.

19. What is the purpose of the weights and biases in a neural network?

Ans. Weights and biases are used in the hidden layers of a neural network to perform computations. The weights are adjusted during the learning process to reduce the error and find the correct output.

20. What kind of data is used for a Regression model, and what kind for a Classification model?

Ans. A Regression model uses and predicts continuous data, such as temperature or price. A Classification model works on discrete data and classifies it into distinct categories or classes, like "hot" or "cold".

3 MARKS QUESTIONS AND ANSWERS

1. Explain the relationship between Artificial Intelligence, Machine Learning, and Deep Learning using the "funnel type approach" described in the document.

Ans. The relationship between AI, ML, and DL can be visualized as a funnel.

Artificial Intelligence is the broadest concept, representing all techniques that enable computers to mimic human intelligence.

Machine Learning is a subset of AI, where machines learn from data and experience.

Deep Learning is the most specific subset of ML, which uses artificial neural networks and vast amounts of data to train itself. This hierarchy forms a funnel, with many applications of AI, a fewer number of ML applications, and a very specific set of DL applications.

2. Describe the concept of Supervised Learning and provide an example from the document.

Ans. Supervised Learning is a machine learning approach where the model is trained on a labeled dataset. This means the data is tagged with some information that acts as a supervisor or teacher, guiding the machine to learn the relationship between features and labels. An example from the document is a model trained to predict the currency of a coin based on its weight. The weights of the coins are the features, and the currency (Euro, Dirham, etc.) is the label. Once trained, the model can identify a new coin based on its weight.

3. How does a Learning-Based Approach differ from a Rule-Based Approach, particularly regarding adaptability?

Ans. The primary difference lies in adaptability. A Rule-Based Approach is static; the machine follows a predefined set of rules and does not learn from its mistakes or adapt to new, unforeseen data. In contrast, a Learning-Based Approach is dynamic and adaptive. The model is trained on data and is able to modify its own algorithms to handle new situations and exceptions, continuously improving its accuracy over time. An example is a learning-based spam filter that adapts to new types of spam emails.

4. Explain the role of the input, hidden, and output layers in a simple Artificial Neural Network.

Ans. An Artificial Neural Network (ANN) consists of three main layers. The input layer is the first layer, responsible for acquiring the data and feeding it into the network without any processing. The hidden layers are where all the computations and processing take place. Each

node in these layers runs its own machine learning algorithm using weights and biases on the data it receives. The output layer is the final layer that receives the fully processed data from the last hidden layer and presents it as the final output to the user, also without any processing.

5. What are the two types of Supervised Learning models, and how do they differ?

Ans. The two types are Classification and Regression. Classification models are used when the output is a discrete or categorical value. They classify data into predefined classes based on the training data. An example is classifying an email as 'spam' or 'not spam'.

Regression models, on the other hand, are used for continuous data and predict a continuous output value. An example is predicting the price of a house or a car.

6. How does Reinforcement Learning work? Use an example from the document to illustrate.

Ans. Reinforcement Learning is a trial-and-error approach where a machine learns to perform a task by making a series of decisions to maximize a reward metric. For example, if you provide an image of an apple and the machine incorrectly predicts "cherry," you give it negative feedback. The machine learns from this mistake. When it later predicts "apple" correctly, you give it positive feedback, and it learns that this is the right answer. The model continuously learns and improves its decisions based on these rewards and punishments.

7. Explain the concept of Unsupervised Learning and its purpose.

Ans. Unsupervised Learning is a type of machine learning that works on unlabeled data. The person training the model may not have any information about the data. The model's purpose is to independently identify relationships, patterns, and trends within the data. It is used to group or "cluster" data points based on similarities, helping the user understand the underlying structure of the data without any prior guidance.

8. In a neural network, how does the process of finding the right output occur?

Ans. The process of finding the right output in a neural network begins with a trial-and-error method. Data is passed from the input layer through the hidden layers, where computations using weights and biases are performed. The output layer then provides a final output. If the output is incorrect, the network calculates the error and adjusts the weights accordingly. This process repeats, with the weights being fine-tuned with each try until the network "learns" and produces the correct or desired output.

9. Describe the two sub-categories of Unsupervised Learning with a brief example for each.

Ans. The two sub-categories are Clustering and Association.

Clustering involves grouping unlabeled data points into clusters based on similarities in their characteristics. An example is an OTT platform grouping songs with a slow tempo and soft intensity into one cluster to recommend to a user who likes similar music.

Association is a method used to find interesting relationships between variables in a database. An example is a supermarket finding a relationship that customers who buy bread are also likely to buy butter, which can be used for product recommendations.

10. How do Machine Learning models improve at tasks with experience?

Ans. Machine Learning models improve at tasks with experience by learning from new data and their own mistakes. The machine takes into account the times it went wrong and considers exceptions in future iterations. This allows the model to improvise and become more accurate over time. It is an adaptive process, unlike the static nature of a rule-based model.

11. Explain the concept of Anomaly Detection as an example of Machine Learning.

Ans. Anomaly Detection is a machine learning technique used to find unexpected or unusual things in data. The model learns the normal patterns from a dataset, and then flags any data point that deviates significantly from that normal pattern as an anomaly. For example, a model

tracking a person's heart rate could identify a sudden, abnormal spike as an anomaly, which could signal a potential health issue.

12. What is the difference between a Regression and a Classification model?

Ans. Both are supervised learning models, but they differ in the type of output they predict. A Regression model predicts a continuous value, which can be any number within a range, such as a temperature or price. A

Classification model, on the other hand, predicts a discrete or categorical value, assigning the data to a specific class or category, such as "hot" or "cold" weather, or "spam" or "not spam" email.

13. Give an example of a supervised learning problem and identify the feature and the label.

Ans. A supervised learning problem is to predict whether a customer is eligible for a bank loan or not. Here, the features would be the parameters of the customer, such as their credit score, income, employment history, etc. The label would be the final decision: "eligible" or "not eligible," which is the target variable that the model is trained to predict.

14. Describe the process of a Learning-Based spam email filter and explain why it is more effective than a Rule-Based one.

Ans. A Learning-Based spam filter is trained with a large dataset of emails, each labeled as either "spam" or "legitimate". The filter analyzes the content and characteristics of these emails to learn patterns that distinguish them. It is more effective than a rule-based one because it continuously adapts and improves its accuracy over time as it encounters new examples of spam. A rule-based filter would only be able to identify spam based on predefined rules, failing to catch new types of spam that don't match those rules.

15. What are the key takeaways from the "Test Yourself" section regarding Supervised vs. Unsupervised Learning?

Ans. The key takeaways are:

- 1) A social media platform identifying a friend in a tagged photo album is supervised learning because the photos are already tagged (labeled).
- 2) An OTT platform recommending content based on watch history is unsupervised learning because the model analyzes patterns in the unlabeled watch history to make recommendations.
- 3) A bank analyzing data for suspicious transactions without a predefined definition of "suspicious" is also unsupervised learning because the model is identifying outliers or anomalous transactions without a "fraud" label.

16. What is a Training Data Set? How is it similar to a teacher-student relationship?

Ans. A training data set is a collection of examples given to a model to analyze and learn from. It is similar to a teacher-student relationship because, just as a teacher uses numerous examples and illustrations to teach a topic to a class, a set of labeled data is used to train an AI model. The model learns from this data before being tested on its own.

17. What is the difference between a feature and a label in a dataset?

Ans. Features are the descriptive attributes or columns of a dataset, such as color, size, and price. A label is a special type of feature that serves as the tag or target variable that the model is trying to predict. The role of a feature versus a label depends on the context of the problem.

18. In the context of the perceptron example, how do the values for weights and bias affect the final decision?

Ans. The values for weights and biases are determined by personal preference and experience. The weights represent the importance of each input factor; a higher weight means the factor is more important. The bias value determines the cautiousness of the decision-making. A higher

bias value makes the decision gear towards a more cautious outcome, while a lower value makes it more daring. Therefore, the choice of these values directly affects the outcome.

19. Describe the concept of Object Identification in Deep Learning and give an example from the document.

Ans. Object identification in deep learning is the task of identifying and labeling objects within an image. It uses powerful algorithms to analyze what is in a picture and categorize those things. The document gives an example of a deep learning model that takes pixels of a bird image as input and is able to correctly analyze and predict that it is a bird using an ANN algorithm.

20. What is a "Convolutional Neural Network" (CNN) and how is it different from a standard ANN?

Ans. A Convolutional Neural Network (CNN) is a type of Deep Learning algorithm. While both ANNs and CNNs are types of neural networks, the document implies that CNNs are specifically designed to process image-based data. They work by assigning importance (learnable weights and biases) to various aspects of an image to differentiate one from another. This makes them highly effective for tasks like object identification and image classification.

5 MARKS QUESTIONS AND ANSWERS

1. Explain the different types of learning-based AI models: Supervised, Unsupervised, and Reinforcement Learning. Discuss their key characteristics, data requirements, and provide a unique example for each.

Ans. **Supervised Learning** is a model that requires a labeled dataset for training. The goal is to predict an output based on the input features. It's like a teacher guiding a student with solved examples. For example, a model trained on a dataset of house features (e.g., number of bedrooms, location, square footage) labeled with their selling prices can predict the price of a new house.

Unsupervised Learning works on unlabeled data, where there is no guidance or supervision. The model's purpose is to discover hidden patterns, relationships, or groups within the data on its own. For example, a geneticist might use an unsupervised learning model to find previously unknown patterns in DNA sequences without any prior knowledge of what to look for.

Reinforcement Learning involves a machine learning by interacting with an environment through a trial-and-error process, receiving rewards for good actions and penalties for bad ones. The goal is to learn a policy that maximizes the total reward over time. For example, a robot learning to navigate a maze would receive a positive reward for moving towards the exit and a negative one for hitting a wall, eventually learning the optimal path.

2. Describe the structure and function of an Artificial Neural Network. Explain how information flows through its layers to produce an output.

Ans. An Artificial Neural Network (ANN) is a computational model inspired by the human brain. It consists of three main types of layers: an input layer, one or more hidden layers, and an output layer.

Input Layer: This is the first layer where the network receives raw data. It does not perform any processing; its sole function is to acquire the data and pass it to the next layer.

Hidden Layers: These are the intermediate layers where the actual computations occur. Each node within a hidden layer has its own algorithm and is connected to nodes in the previous and next layers. Data from the input layer is processed here using weights (which determine the importance of the input) and biases (which adjust the output). The number of hidden layers and nodes depends on the complexity of the problem.

Output Layer: This is the final layer that receives the processed information from the last hidden layer. Like the input layer, it does not perform any computation. It simply provides the final output to the user. The entire process is a continuous loop of trial and error, where weights are adjusted based on the error to improve accuracy.

3. Explain the concept of the Rule-Based Approach and its key characteristics. Why is it considered a static form of AI, and what are its limitations?
Ans. The **Rule-Based Approach** is a type of AI modeling where the developer explicitly defines the rules and patterns that the machine must follow. The machine reacts according to these predefined instructions to deliver the desired output. A classic example is a chatbot that uses a decision tree with rules like "If the user asks about order status, ask for their order number". This approach is considered static because the learning is fixed; the model, once trained, does not adapt to changes in the original dataset or learn from its mistakes. If it encounters a scenario that doesn't fit a defined rule, it will likely fail and will not improvise. This is a major limitation, as it lacks the adaptability of learning-based models and cannot handle exceptions or unforeseen circumstances effectively.
4. Compare and contrast Classification and Regression models. Give an example of a real-world scenario for each and explain why one model is more suitable than the other for that scenario.
Ans. Both Classification and Regression are sub-categories of Supervised Learning, but they differ fundamentally in the type of output they predict.
Classification: This model predicts a discrete or categorical output. It is used when the answer falls into a predefined class.
Scenario: A bank wants to predict if a customer will default on their loan or not. The output is a binary choice: "Yes" or "No." A classification model is suitable because the task is to assign the customer to one of two distinct categories.
Regression: This model predicts a continuous value, which can be any number within a range.
Scenario: A real estate company wants to predict the selling price of a house. The output is a continuous numerical value (e.g., \$350,000, \$425,500). A regression model is suitable because the task is to predict a specific, non-discrete number based on various input features like the number of rooms and location.
5. Describe the concept of Unsupervised Learning and its two main sub-categories: Clustering and Association. Explain how they are used to find patterns in data.
Ans. **Unsupervised Learning** is a type of learning that operates on unlabeled data to find hidden patterns and relationships without any guidance. The model is responsible for discovering these patterns on its own. Its two main sub-categories are:
Clustering: The goal of clustering is to group data points into "clusters" based on their similarities. The model identifies shared characteristics and organizes the data into meaningful groups. For example, a streaming service like Netflix uses clustering to group users with similar viewing habits and then recommends content based on what others in their cluster have watched.
Association: This method is used to find interesting relationships or "association rules" between variables in a dataset. It finds items that tend to occur together. A well-known example is a supermarket analyzing transaction data to find that customers who buy diapers also tend to buy baby wipes. This information can be used for store layout optimization or targeted marketing.
6. Explain how a perceptron, a simplified model of a neural network, makes a decision. Use the example of deciding whether to go out or not.
Ans. The perceptron is a fundamental concept that illustrates how an AI makes a decision. It takes several inputs, each representing a factor that influences the decision. In the example of deciding whether to go out, the inputs could be: "Do I have a jacket?", "Is it sunny now?", "Do I have an umbrella?", and "What is the weather forecast?". Each input is assigned a weight that represents its importance. For instance, "Is it sunny now?" might have a higher weight than "Do I have an umbrella?". A bias value is also added, which reflects the individual's cautiousness. The perceptron then sums up the weighted inputs and the bias, and compares this total to a threshold. If the sum exceeds the threshold, the output is "Yes, go out." If not, the output is "No,

stay in." The values for weights and bias are learned from experience and preference, which is why different people can make different decisions in the same situation.

7. What are the key concepts and terminologies related to data that are essential for understanding AI modeling? Explain the role of each.

Ans. Understanding key data terminologies is crucial for AI modeling.

Data: Data is information in any form, such as a table of information about fruits. It is the raw material that fuels AI models.

Features: These are the descriptive attributes of the data, represented by the columns of a table. For example, in a fruit dataset, "Fruit", "Color", and "Price" are features.

Labels: A label is a special feature that provides a tag or meaning to the data. It is the target variable that the model is trained to predict. In a fruit example, if the model predicts the fruit name based on color, the fruit name is the label.

Training Dataset: This is the collection of labeled data used to teach the model to analyze and learn.

Testing Dataset: This dataset, which may or may not be labeled, is used to test the accuracy and performance of the trained model.

8. Explain the concept of Reinforcement Learning and discuss why it is particularly important for handling large complex problems where pre-existing data may be insufficient.

Ans. Reinforcement Learning (RL) is a learning approach where an agent learns to make a sequence of decisions in an environment to maximize a cumulative reward. It learns through a process of trial and error, receiving positive feedback (rewards) for desired actions and negative feedback (penalties) for undesired ones.

RL is particularly important for large, complex problem spaces because it does not require a lot of pre-existing knowledge or data to provide useful solutions. In situations where the environment is unforeseen or constantly changing, and there isn't enough data for supervised or unsupervised learning, an RL model can be adaptive and learn the optimal strategy by exploring and exploiting the environment. For example, a model learning to park a car through RL doesn't need a labeled dataset of parking scenarios; it learns by trying different maneuvers and adjusting its actions based on the feedback it receives (e.g., a reward for getting closer to the parking spot and a penalty for hitting a curb).

9. Imagine a healthcare organization is using an AI model for two different tasks related to patient hospital stays.

a) To predict whether a patient will have a short or long hospital stay.

b) To predict the number of days a patient will stay in the hospital.

Identify the type of model (classification or regression) that would be suitable for each task and justify your choice.

Ans. a) Predicting a short or long hospital stay: This task requires a Classification model. A classification model is used when the output is a discrete category or class. The model would be trained on patient data and learn to classify each patient into one of two predefined categories: "short stay" or "long stay." The output is not a continuous value but a categorical one.

b) Predicting the number of days a patient will stay: This task requires a Regression model. A regression model is used when the output is a continuous value. The model would be trained to predict a specific number (e.g., 3 days, 7 days, 15 days), which is a continuous variable. The document explicitly states that regression models predict continuous values like age or temperature.

10. What is the difference between supervised and unsupervised learning, and how do they relate to the concepts of "teacher and student" and "learning without guidance"?

Ans. The primary difference between supervised and unsupervised learning is the presence of a supervisor or guidance during the training process.

Supervised Learning is analogous to a teacher and student relationship. The model is the student, and the labeled data acts as the teacher, providing solved examples for the student to learn from. The model learns the relationship between the features and the labels and then applies this knowledge to new, unseen data. The goal is to predict a known outcome.

Unsupervised Learning is like learning without any guidance. The model is given an unlabeled dataset and must discover patterns, relationships, and trends on its own. There is no "correct" answer provided. The model's goal is to find the inherent structure within the data, such as grouping similar data points together in clusters. The analogy provided in the document is a child learning to swim on their own without supervision, discovering the techniques by themselves.

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